Bayesian Compressive Sensing Using Normal Product Priors

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***Abstract*—In this letter, we introduce a new sparsity-promoting prior, namely, the "normal product" prior, and develop an efficient algorithm for sparse signal recovery under the Bayesian framework. The normal product distribution is the distribution of a product of two normally distributed variables with zero means and possibly different variances. Like other sparsity-encouraging distributions such as the Student's t-distribution, the normal product distribution has a sharp peak at the origin, which makes it a suitable prior to encourage sparse solutions. A two-stage normal product-based hierarchical model is proposed. We resort to the variational Bayesian (VB) method to perform the inference. Simulations are conducted to illustrate the effectiveness of our proposed algorithm as compared with other state-of-the-art compressed sensing algorithms.**

***Index Terms*—Compressed Sensing, normal product prior, sparse Bayesian learning.**

I. Introduction

COMPRESSED sensing [1], [2] is a new data acquisition technique that has attracted much attention over the past decade. Existing methods for compressed sensing can generally be classified into the following categories, i.e. the greedy pursuit approach [3], the convex relaxation-type approach [4] and the nonconvex optimization method [5]. Another class of compressed sensing techniques that have received increasing attention are Bayesian methods. In the Bayesian framework, the signal is usually assigned a sparsity-encouraging prior, such as the Laplace prior and the Gaussian-inverse Gamma prior [6], to encourage sparse solutions. It has been shown in a series of experiments that Bayesian compressed sensing techniques [7] demonstrate superiority for sparse signal recovery as compared with the greedy methods and the basis pursuit method. One of the most popular prior model for Bayesian compressed sensing is the Gaussian-inverse Gamma prior proposed in [6]. The Gaussian-inverse Gamma prior is a two-layer hierarchical model in which the first layer specifies a Gaussian prior to the sparse signal and an inverse Gamma priori is assigned to the parameters characterizing the Gaussian prior. As discussed in [8], this two-stage hierarchical model is equivalent to imposing a Student's t-distribution on the sparse signal. Besides the Gaussian-inverse Gamma prior, authors in [9] employ Laplace priors for the sparse signal to promote sparse solutions.

In this letter, we introduce a new sparsity-encouraging prior, namely, the normal product (NP) prior, for sparse signal recovery. A two-stage normal product-based hierarchical model is established, and we resort to the variational Bayesian (VB) method to perform the inference for the hierarchical model. Our experiments show that the proposed algorithm achieves similar performance as the sparse Bayesian learning method, while with lower computational complexity.

Manuscript received July 26, 2014; accepted October 16, 2014. Date of publication October 21, 2014; date of current version November 03, 2014. This work was supported in part by the National Science Foundation of China under Grant 61172114, and by the Sichuan Province Science and Technology Program under Grant 2013GZ0029. The associate editor coordinating the review of this manuscrpt and approving it for publication was Prof. Alex Dimakis.

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Digital Object Identifier 10.1109/LSP.2014.2364255

II. The Bayesian Network

In the context of compressed sensing, we are given a noise corrupted linear measurements of a vector $\mathbf{x}_0$

$$\mathbf{y} = \mathbf{A}\mathbf{x}_0 + \mathbf{n} \tag{1}$$

Here $\mathbf{A} \in \mathbf{R}^{M\times N}$ is an underdetermined measurement matrix with low coherence of columns, $\mathbf{n} \in \mathbf{R}^M$ represents the acquisition noise vector and $\mathbf{x}_0$ a sparse signal. For the inverse problem, both $\mathbf{x}_0$ and $\mathbf{n}$ are unknown, and our goal is to recover $\mathbf{x}_0$ from $\mathbf{y}$. We formulate the observation noise as zero mean independent Gaussian variable with variance ξ^2 i.e. $\mathbf{y} \sim N(\mathbf{A}\mathbf{x}_0, \xi^2\mathbf{I})$ and seek a sparse estimation $\mathbf{x} \triangleq (x_1, x_2, \ldots x_N)$ for $\mathbf{x}_0$.

In this section, we utilize a two-layer hierarchical Bayesian prior model for the signal. In the first stage of prior, we use the normal product (NP) distribution as sparseness prior:

$$\mathbf{x} \sim NP(\mathbf{0}, \mathbf{\Sigma})$$

where $\mathbf{0} \in \mathbf{R}^N$ stands for its mean and $\mathbf{\Sigma}$ given as a diagonal matrix $\mathrm{diag}({\sigma_1}^2, {\sigma_2}^2, \ldots, {\sigma_N}^2) \in \mathbf{R}^{N\times N}$ represents its covariance matrix. For each element x_i of $\mathbf{x}$, its probability density distribution is $K_0(\frac{|x_i|}{\sigma_i})/\pi\sigma_i$, which exhibits a sharp peak at the origin and heavy tails, where $K_0(\cdot)$ is the zero order modified Bessel function of the second kind [10] thus the probability density function (PDF) of $\mathbf{x}$ is

$$\frac{1}{\pi^N}\prod_{i=1}^{N}\frac{1}{\sigma_i}K_0\left(\frac{|x_i|}{\sigma_i}\right). \tag{2}$$

A NP distributed scaler variable $x_i \in \mathbf{R}$ can be decomposed into the product of two independent normally distributed variables i.e. if $x_i \sim NP(0, \sigma_i^2)$, then there are $a_i \sim N(0, \kappa_i^2)$ and $b_i \sim N(0, \gamma_i^2)$ satisfying $x_i = a_i b_i$ with moment relationship $\sigma_i = \gamma_i\kappa_i$[11]. We call this property as the generating rule of normal product distribution. Similarly for the vector $\mathbf{x}$, we can

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decompose it into the Hadamard product of two independent virtual normally distributed vector variables $\mathbf{a}$ and $\mathbf{b}$ whose covariance matrix are diagonal matrix $\boldsymbol{\kappa}^2 \triangleq \mathrm{diag}(\kappa_1^2, \kappa_2^2, \ldots, \kappa_N^2)$ and $\boldsymbol{\gamma}^2 \triangleq \mathrm{diag}(\gamma_1^2, \gamma_2^2, \ldots, \gamma_N^2)$ respectively i.e. $\mathbf{x} = \mathbf{a} \circ \mathbf{b}$ and $\boldsymbol{\Sigma} = \boldsymbol{\kappa}^2 \circ \boldsymbol{\gamma}^2$, where $\circ$ denotes the Hadamard product. Finally, we set σ_i^{-2} as a realization of Gamma hyperprior $\Gamma(\alpha, \beta)$ and choose $\alpha \to 0$, $\beta \to 0$ to construct a sharper distribution of $\mathbf{x}$ during the variational procedure [12] for the second stage of the signal model.

According to the generating rule of normal product mentioned above, the two parallel normally distributed variables $\mathbf{a}$ and $\mathbf{b}$ can be introduced as the hidden variables. By integrating out it, the posterior distribution can be expressed as:

$$P_{\mathbf{x}}(\mathbf{x}, \boldsymbol{\Sigma}|\mathbf{y}; \alpha, \beta) = \int P(\mathbf{x}, \mathbf{a}, \mathbf{b}, \boldsymbol{\Sigma}|\mathbf{y}; \alpha, \beta) d\mathbf{a} d\mathbf{b}$$

where

$$\begin{aligned} &P(\mathbf{x}, \mathbf{a}, \mathbf{b}, \boldsymbol{\Sigma}|\mathbf{y}; \alpha, \beta) \\ &\propto P(\mathbf{y}|\mathbf{x}) P(\mathbf{x}|\mathbf{a}, \mathbf{b}) P(\mathbf{a}|\boldsymbol{\kappa}^2) P(\mathbf{b}|\boldsymbol{\gamma}^2) P(\boldsymbol{\Sigma}|\alpha, \beta) \\ &= P(\mathbf{y} - \mathbf{A}\mathbf{x}) \delta(\mathbf{x} - \mathbf{a} \circ \mathbf{b}) P(\mathbf{a}|\boldsymbol{\kappa}^2) P(\mathbf{b}|\boldsymbol{\gamma}^2) P(\boldsymbol{\Sigma}|\alpha, \beta) \end{aligned}$$

and $\delta(\cdot)$ denotes the Dirac Delta function. In this expression, we know that the value of $\mathbf{x}$ must be equal to $\mathbf{a} \circ \mathbf{b}$ while keeping the value of $P(\mathbf{x}, \mathbf{a}, \mathbf{b}, \boldsymbol{\Sigma}|\mathbf{y}; \alpha, \beta)$ nonzero. Consider the Bayesian Risk function:

$$\begin{aligned} E(L(\hat{\mathbf{x}}, \mathbf{x})) &= \int L(\hat{\mathbf{x}}, \mathbf{x}) P_{\mathbf{x}}(\mathbf{x}, \boldsymbol{\Sigma}|\mathbf{y}; \alpha, \beta) d\mathbf{x} d\boldsymbol{\Sigma} \\ &\propto \int L(\hat{\mathbf{x}}, \mathbf{a} \circ \mathbf{b}) P_{\mathbf{a},\mathbf{b}}(\mathbf{a}, \mathbf{b}, \boldsymbol{\Sigma}|\mathbf{y}; \alpha, \beta) d\mathbf{a} d\mathbf{b} d\boldsymbol{\Sigma} \end{aligned}$$

where

$$\begin{aligned} &P_{\mathbf{a},\mathbf{b}}(\mathbf{a}, \mathbf{b}, \boldsymbol{\Sigma}|\mathbf{y}; \alpha, \beta) \\ &= P(\mathbf{y} - \mathbf{A}(\mathbf{a} \circ \mathbf{b})) P(\mathbf{a}|\boldsymbol{\kappa}^2) P(\mathbf{b}|\boldsymbol{\gamma}^2) P(\boldsymbol{\Sigma}|\alpha, \beta) \end{aligned}$$

and $L(\hat{\mathbf{x}}, \mathbf{x})$ represents the loss function. Thus, we can replace $P_{\mathbf{x}}(\mathbf{x}, \boldsymbol{\Sigma}|\mathbf{y}; \alpha, \beta)$ with $P_{\mathbf{a},\mathbf{b}}(\mathbf{a}, \mathbf{b}, \boldsymbol{\Sigma}|\mathbf{y}; \alpha, \beta)$ in the inference procedure while maintaining the value of the Bayesian risk function at a consistent level. So the modified Bayesian Network is:

$$\mathbf{y} \sim N(\mathbf{A}(\mathbf{a} \circ \mathbf{b}), \xi^2 \mathbf{I}) \quad (3)$$
$$\mathbf{a} \sim N(\mathbf{0}, \boldsymbol{\kappa}^2) \quad (4)$$
$$\mathbf{b} \sim N(\mathbf{0}, \boldsymbol{\gamma}^2) \quad (5)$$
$$\sigma_i^{-2} = \gamma_i^{-2} \kappa_i^{-2} \sim \Gamma(\alpha, \beta) \quad (6)$$

as depicted in Fig. 2.

III. The Varational Bayesian Inference

In order to infer the bayesian network, we use the mean-field variational Bayesian method to analytically approximate the entire posterior distribution of latent variables. In this theorem, it was assumed that the approximate variational distribution $q(\mathbf{z}|\mathbf{y})$ where $\mathbf{z}$ stands for the latent variable in the model could be factorized into the product $\prod_{i=1}^{l} q_i(\mathbf{z}_i|\mathbf{y})$ for the partition $\mathbf{z}_1, \ldots, \mathbf{z}_l$ of $\mathbf{z}$. It could be shown that the solution of the variational method for each factor $q_i(\mathbf{z}_i|\mathbf{y})$ could be written as:

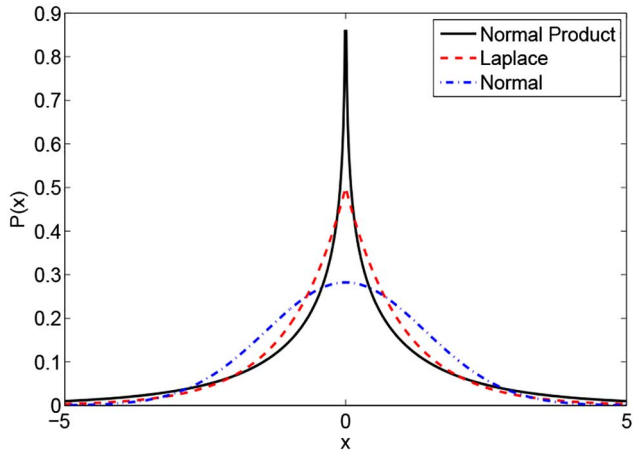


Fig. 1. Three kinds of PDF with the same standard deviation.

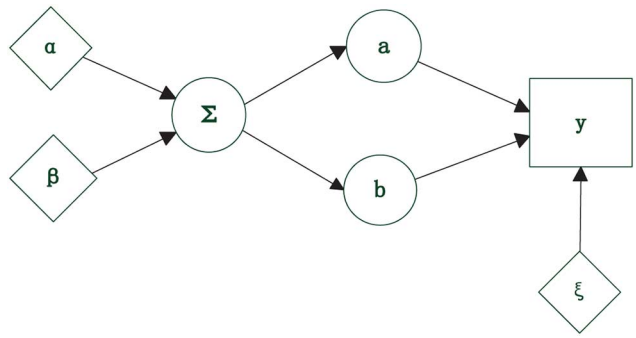


Fig. 2. The Bayesian network.

$$q_i^*(\mathbf{z}_i|\mathbf{y}) = \frac{e^{<\ln p(\mathbf{z}|\mathbf{y})>_{\neq \mathbf{z}_i}}}{\int e^{<\ln p(\mathbf{z}|\mathbf{y})>_{\neq \mathbf{z}_i}} d\mathbf{z}_i}$$

where $< \ln p(\mathbf{z}|\mathbf{y}) >_{\neq \mathbf{z}_i}$ means taking the expectation over the set of $\mathbf{z}$ using the variational distribution except $\mathbf{z}_i$[13]. Applying the variational Bayesian method upon the Bayesian model mentioned in Section II, the posterior approximation of $\boldsymbol{\Sigma}, \mathbf{a}, \mathbf{b}$ are respectively:

$$\ln(q^*(\mathbf{a})) \propto < \ln P(\mathbf{a}, \mathbf{b}, \boldsymbol{\Sigma}|\mathbf{y}; \alpha, \beta) >_{q^*(\mathbf{b}), q^*(\boldsymbol{\Sigma})} \quad (7)$$
$$\ln(q^*(\mathbf{b})) \propto < \ln P(\mathbf{a}, \mathbf{b}, \boldsymbol{\Sigma}|\mathbf{y}; \alpha, \beta) >_{q^*(\mathbf{a}), q^*(\boldsymbol{\Sigma})} \quad (8)$$
$$\ln(q^*(\boldsymbol{\Sigma})) \propto < \ln P(\mathbf{a}, \mathbf{b}, \boldsymbol{\Sigma}|\mathbf{y}; \alpha, \beta) >_{q^*(\mathbf{b}), q^*(\mathbf{a})} \quad (9)$$

A. Posterior Approximation of $\mathbf{a}$ and $\mathbf{b}$

Substituting (3), (4), (5) and (6) into (8), we can arrive at:

$$q^*(\mathbf{b}) \sim N(\boldsymbol{\Gamma}'^{-1}\mathbf{c}, \boldsymbol{\Gamma}'^{-1})$$

where

$$\boldsymbol{\Gamma}' = \frac{1}{\xi^2}(\mathbf{A}^T\mathbf{A}) \circ < \mathbf{a}\mathbf{a}^T >_{q^*(\mathbf{a})} + < \boldsymbol{\gamma} >_{q^{**}(\boldsymbol{\gamma})}^{-2}$$

and

$$\mathbf{c}' = \frac{1}{\xi^2} \mathrm{diag}\left(< \mathbf{a} >_{q^*(\mathbf{a})} \right) \mathbf{A}^T \mathbf{y}.$$

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From the principle of variational inference, we know that $q(\mathbf{b})$ is an approximation of $P(\mathbf{b}|\mathbf{y})$ i.e. $P(\mathbf{b}|\mathbf{y}) \approx q(\mathbf{b})$ and in order to get a more concise iteration formula, we choose the relaxation of the secondary moment as:

$$< \mathbf{a}\mathbf{a}^T >_{q^*(\mathbf{a})} \approx < \mathbf{a} >_{q^{**}(\mathbf{a})} < \mathbf{a}^T >_{q^{**}(\mathbf{a})}$$
$$< \boldsymbol{\gamma}^{-2} >_{q^*(\boldsymbol{\gamma})} \approx < \boldsymbol{\gamma} >^{-2}_{q^{**}(\boldsymbol{\gamma})}$$

This relaxation can be interpreted as ignoring the posterior variance during the updating, which is inspired by the fact that after the learning procedure the posterior variance always approaches zero to ensure the posterior mean's concentration on the estimated value.

Then the corrected posterior approximation is $q^{**}(\mathbf{b}) \sim N(\boldsymbol{\Gamma}^{-1}\mathbf{c}, \boldsymbol{\Gamma}^{-1})$ where

$$\boldsymbol{\Gamma}^{-1} = \{\frac{1}{\xi^2}\left(\mathbf{A}\mathrm{diag}\left(< \mathbf{a} >_{q^{**}(\mathbf{a})}\right)\right)^T \mathbf{A}\mathrm{diag}\left(< \mathbf{a} >_{q^{**}(\mathbf{a})}\right) + < \boldsymbol{\gamma} >^{-2}_{q^{**}(\boldsymbol{\gamma})}\}^{-1}$$

and

$$\mathbf{c} = \frac{1}{\xi^2}\mathrm{diag}\left(< \mathbf{a} >_{q^{**}(\mathbf{a})}\right)\mathbf{A}^T\mathbf{y}$$

In the noise free case, using Woodbury identity we have:

$$\lim_{\xi\to 0} \boldsymbol{\Gamma}^{-1}\mathbf{c} = < \boldsymbol{\gamma} >_{q^{**}(\boldsymbol{\gamma})} (\mathbf{A}\mathrm{diag}(< \mathbf{a} >_{q^{**}(\mathbf{a})}) < \boldsymbol{\gamma} >_{q^{**}(\boldsymbol{\gamma})})^+ \mathbf{y} \tag{10}$$

and

$$\lim_{\xi\to 0} \boldsymbol{\Gamma}^{-1} = (\mathbf{I} - < \boldsymbol{\gamma} >_{q^{**}(\boldsymbol{\gamma})} \left(\mathbf{A}\mathrm{diag}\left(< \mathbf{a} >_{q^{**}(\mathbf{a})}\right) < \boldsymbol{\gamma} >_{q^{**}(\boldsymbol{\gamma})}\right)^+) \times \mathbf{A}\mathrm{diag}\left(< \mathbf{a} >_{q^{**}(\mathbf{a})}\right) < \boldsymbol{\gamma} >^2_{q^{**}(\boldsymbol{\gamma})} \tag{11}$$

Similarly, for $\mathbf{a}$, substituting (3), (4), (5) and (6) into (7) we have:

$$q^*(\mathbf{a}) \sim N(\mathbf{H}'^{-1}\mathbf{f}', \mathbf{H}'^{-1})$$

where

$$\mathbf{H}' = \frac{1}{\xi^2}(\mathbf{A}^T\mathbf{A}) \circ < \mathbf{b}\mathbf{b}^T >_{q^*(\mathbf{b})} + < \boldsymbol{\kappa}^{-2} >_{q^*(\boldsymbol{\kappa})}$$

and

$$\mathbf{f}' = \frac{1}{\xi^2}\mathrm{diag}\left(< \mathbf{b} >_{q^*(\mathbf{b})}\right)\mathbf{A}^T\mathbf{y}$$

So the corrected approximation is:

$$q^{**}(\mathbf{a}) \sim N(\mathbf{H}^{-1}\mathbf{f}, \mathbf{H}^{-1})$$

Then again, in the noiseless case, we have

$$\lim_{\xi\to 0} \mathbf{H}^{-1}\mathbf{f} = < \boldsymbol{\kappa} >_{q^{**}(\boldsymbol{\kappa})} (\mathbf{A}\mathrm{diag}(< \mathbf{b} >_{q^{**}(\mathbf{b})}) < \boldsymbol{\kappa} >_{q^{**}(\boldsymbol{\kappa})})^+ \mathbf{y} \tag{12}$$

and

$$\lim_{\xi\to 0} \mathbf{H}^{-1} = (\mathbf{I} - < \boldsymbol{\kappa} >_{q^{**}(\boldsymbol{\kappa})} \left(\mathbf{A}\mathrm{diag}\left(< \mathbf{b} >_{q^{**}(\mathbf{b})}\right) < \boldsymbol{\kappa} >_{q^{**}(\boldsymbol{\kappa})}\right)^+) \times \mathbf{A}\mathrm{diag}\left(< \mathbf{b} >_{q^{**}(\mathbf{b})}\right) < \boldsymbol{\kappa} >^2_{q^{**}(\boldsymbol{\kappa})} \tag{13}$$

B. Posterior Approximation of $\boldsymbol{\Sigma}$

As discussed in Section II, we have:

$$\ln(q^*(\boldsymbol{\Sigma})) = \ln(q^*(\boldsymbol{\kappa}^2 \circ \boldsymbol{\gamma}^2)) = \ln(q^*(\boldsymbol{\kappa}^2)) + \ln(q^*(\boldsymbol{\gamma}^2))$$
$$\propto < \ln P(\mathbf{a}, \mathbf{b}, \boldsymbol{\kappa}^2 \circ \boldsymbol{\gamma}^2 | \mathbf{y}; \alpha, \beta) >_{q^{**}(\mathbf{b}), q^{**}(\mathbf{a})}$$

Substituting (4), (5), (6) into (9) and using the separability of $\ln(q^*(\boldsymbol{\kappa}^2))$, it can be shown that:

$$lnq^*\left(\kappa_i^2\right) \propto -\frac{1}{2}\left(< a_i^2 >_{q^{**}(\mathbf{a})} + 2\beta < {\gamma_i}^{-2} >_{q^*(\gamma_i^2)}\right){\kappa_i}^{-2} + (\alpha + \frac{1}{2} - 1)\ln\left({\kappa_i}^{-2}\right)$$

which means

$${\kappa_i}^{-2} \sim \Gamma(\frac{1}{2} + \alpha, \frac{1}{2}(< a_i^2 >_{q^{**}(\mathbf{a})} + 2\beta < {\gamma_i}^{-2} >_{q^*(\gamma^2)}))$$

and

$$< {\kappa_i}^{-2} >_{q^*(\boldsymbol{\kappa}^2)} = \frac{\frac{1}{2} + \alpha}{\frac{1}{2}(< a_i^2 >_{q^{**}(\mathbf{a})} + 2\beta < {\gamma_i}^{-2} >_{q^*(\gamma^2)})} \tag{14}$$

Similarly, we have:

$${\gamma_i}^{-2} \sim \Gamma(\frac{1}{2} + \alpha, \frac{1}{2}(< b_i^2 >_{q^{**}(\mathbf{b})} + 2\beta < {\kappa_i}^{-2} >_{q^*(\boldsymbol{\kappa}^2)}))$$

and

$$< {\gamma_i}^{-2} >_{q^*(\boldsymbol{\gamma}^2)} = \frac{\frac{1}{2} + \alpha}{\frac{1}{2}(< b_i^2 >_{q^{**}(\mathbf{b})} + 2\beta < {\kappa_i}^{-2} >_{q^*(\boldsymbol{\kappa}^2)})} \tag{15}$$

C. Proposed Algorithms

As a result, we use terms "NP-0" and "NP-1" to denote the inference procedure of the one layer signal model and the two layer signal model respectively. The difference between NP-0 and NP-1 is whether the learning process updates precision of normal product prior.

Algorithm 1: NP-0

Input: $\mathbf{A}$, $\mathbf{y}$

Output: $\mathbf{x}^{(t)}$

1: $< \mathbf{a} >^{(0)} \leftarrow \mathbf{1}$, $< \mathbf{b} >^{(0)} \leftarrow \mathbf{1}$, $< \boldsymbol{\gamma} > \leftarrow \mathbf{1}$, $< \boldsymbol{\kappa} > \leftarrow \mathbf{1}$, ε, t_{max},

2: **while** $\|\mathbf{x}^{(t)} - \mathbf{x}^{(t-1)}\| / \|\mathbf{x}^{(t-1)}\| > \varepsilon$ and $t < t_{max}$ **do**

3: Update $< \mathbf{a} >^{(t)}$ using (12)

4: Update $< \mathbf{b} >^{(t)}$ using (10)

5: Update $\mathbf{x}^{(t)} = < \mathbf{a} >^{(t)} \circ < \mathbf{b} >^{(t)}$

6: $t = t + 1$

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7: **end while**

The algorithm NP-0 looks like the FOCUSS algorithm [14], but one of the most fundamental distinctions between them is that NP-0 uses $< \mathbf{a} >^{(t)}$ and $< \mathbf{b} >^{(t)}$, the decomposed component of $\mathbf{x}^{(t)}$, in an interleaved way to update $\mathbf{x}^{(t+1)}$ while FOCUSS uses the whole $\mathbf{x}^{(t)}$ to directly update $\mathbf{x}^{(t+1)}$. In Section IV, we experimentally illustrate that the reconstruction performance of NP-0 is comparable to basis pursuit (BP)[4] which outperforms the basic FOCUSS [8]. Furthermore, we set the initial value of the algorithm as a constant to avoid the local minimum results being returned.

Algorithm 2: NP-1

Input: $\mathbf{A}$, $\mathbf{y}$

Output: $\mathbf{x}^{(t)}$

1: $< \mathbf{a} >^{(0)} \leftarrow \mathbf{1}, < \mathbf{b} >^{(0)} \leftarrow \mathbf{1}, \alpha \leftarrow 0, \beta \leftarrow 0, \varepsilon, t_{max}$

2: **while** $\|\mathbf{x}^{(t)} - \mathbf{x}^{(t-1)}\|/\|\mathbf{x}^{(t-1)}\| > \varepsilon$ and $t < t_{max}$ **do**

3: Update $< \mathbf{a} >^{(t)}$ using (12)

4: Update $Var(\mathbf{a})^{(t)}$ using (13)

5: Update $< \mathbf{b} >^{(t)}$ using (10)

6: Update $Var(\mathbf{b})^{(t)}$ using (11)

7: $\mathbf{x}^{(t)} =< \mathbf{a} >^{(t)} \circ < \mathbf{b} >^{(t)}$

8: Update $< {\kappa_i}^{-2} >^{(t)}_{q^*(\kappa^2)}$ using (14)

9: Update $< {\gamma_i}^{-2} >^{(t)}_{q^*(\gamma^2)}$ using (15)

10: $t = t + 1$

11: **end while**

In the algorithm NP-1, two sparse Bayesian learning (SBL) [6], [8] procedures are coupled to each other and we will experimentally prove that the mean squared error (MSE) via NP-1 descends faster than SBL in Section IV.

IV. Numerical Results

In this section, we compare our proposed algorithms among SBL, iterative reweighted least squares (IRLS)[15] and BP. In the following results, we set the dimension of the original signal to be 100 and in each experiment every entry of the sensing matrix $\mathbf{A}$ is uniformly distributed as standard normal random variable. We ignore the observation noise and only consider the algorithm performance in the noise free case. The stop criterion parameters ε and t_{max} in both algorithm NP-0 and NP-1 are set as 10^{-3} and 300 respectively.

The success rate in Fig. 3 is defined by the average probability of successful restoration, which is counted as the ratio between the number of successful trials and the total number of the independent experiments. It collects the result of 300 Monte Carlo trials and we assume successful trial if $\|\hat{\mathbf{x}} - \mathbf{x}\|_2/\|\mathbf{x}\|_2 < 10^{-3}$, where $\hat{\mathbf{x}}$ is the estimate of the original signal $\mathbf{x}$. MSE in Fig. 4 which measures the average of the squares of the difference between the estimator and estimated signal is calculated as follow:

$$MSE(dB) = 20\log(\|\hat{\mathbf{x}} - \mathbf{x}\|_2/\|\mathbf{x}\|_2)$$

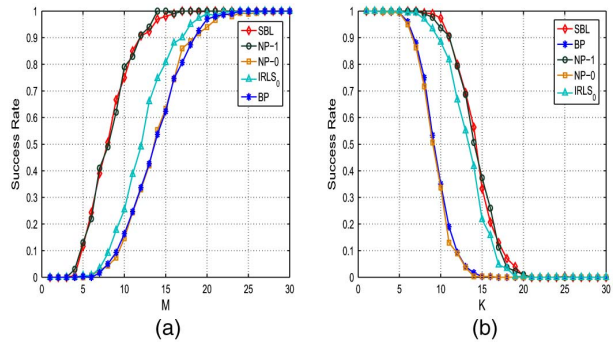


Fig. 3. Simulation results of success rate. (a) $\mathrm{K} = 3$, (b) $\mathrm{M} = 30$.

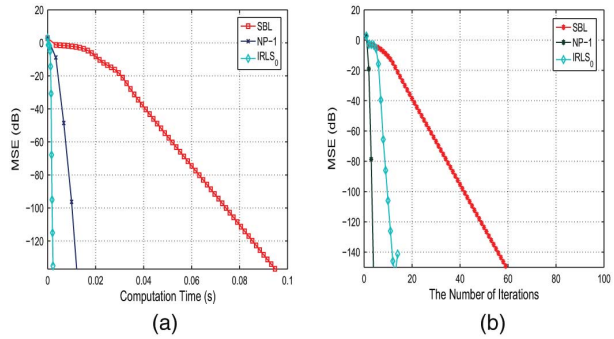


Fig. 4. Simulation results of computational cost. (a) $\mathrm{K} = 3, \mathrm{M} = 30$. (b) $\mathrm{K} = 3, \mathrm{M} = 30$.

Fig. 3 demonstrates the superior performance of the NP-1 algorithm in a few measurements. It shows the success rate of the respective algorithms versus the number of measurement M and sparse level K of the signal. We can see that the reconstruction performance of the one layer normal product algorithm(NP-0) is almost the same as BP, which is the MAP estimation of unknowns in the one layer Laplace prior framework. The comparisons also show that the two layers normal product algorithm(NP-1) requires as few number of CS measurements as SBL while inheriting the similar reconstruction precision.

In Fig. 4, the comparison among the computational cost of the three algorithms on the reconstruction performance is performed on a personal PC with dual-core 3.1 GHz CPU and 4 GB RAM. It is interesting to note that the MSE of NP-1 descends faster than SBL in Fig. 4(a) while it has the same sparsity-undersampling tradeoff performance as SBL. It should also be noticed that we haven't used any basis pruning technology as in [16] to reduce the computational complexity. To understand the reason of low computational time, from Fig. 4(b), we observe that the NP-1 requires only a few numbers of iterations to arrive at the stop condition. In summary, these experimental results confirm that the NP-1 is a fast sparse Bayesian algorithm.

V. Conculsion

In this letter, we formulated a normal product prior based Bayesian framework to solve the compressed sensing problem in the noise free case. Using this framework, we derived two algorithms, namely, NP-0 and NP-1 and compared them with different algorithms. We had shown that the reconstruction performance of our algorithm NP-1 is similar to SBL while the interleaved updating procedure provides improved performance in terms of running times.

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